Sudhanshu Mukherjee

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EXPERIENCE SUMMARY

- Product-oriented Data Scientist and ML Engineer with 4+ years of applied ML life-cycle (data engineering, model development, model serving (REST APIs), CI/CD, and monitoring) and a strong bias for measurable business impact for consumer and enterprise products
- Strengths include LLM & RAG systems, MLflow workflows, Docker/Kubernetes deployments, and translating models into measurable business outcomes.
- Open to **Relocation** with/without assistance, can work without sponsorship through August 2028

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C, Bash
- **ML / DL:** scikit-learn, XGBoost, CatBoost, LightGBM, PyTorch, transformers, LLaVA, LoRA, fine-tuning
- **LLM / RAG:** LangChain, LangGraph, FAISS/vector stores, Azure OpenAI / Google Gemini integrations
- **Big Data & MLOps:** PySpark, Databricks, MLflow, wandb, feature stores, Delta Lake
- **Cloud & Infra:** Azure (Databricks, Azure AI Studio), AWS (Sagemaker, S3, Redshift), Docker, Kubernetes
- **Monitoring & CI/CD:** Git, GitHub Actions / Azure DevOps, Prometheus
- **Data & Viz:** Pandas, Polars, NumPy, SQL optimization, Power BI
- **Other:** Streamlit, Gradio, ML experimentation, A/B testing, causal inference

PROFESSIONAL EXPERIENCE

Data Scientist, Community Dreams Foundation, Remote, Sep 2025 – present

- Built an AI health agent using LangGraph to orchestrate multi-step RAG workflows (ingest, normalize, retrieve, LLM-assisted synthesis, action), enabling personalized recommendations and automated report generation for end users.
- Developed a comprehensive data analysis pipeline using Python and Pandas to ingest, synthesize, and analyze disparate biomedical data streams, including wearables, structured labs, genomics, and user surveys
- Designed a rule-based algorithm to calculate a holistic health score and integrated a Google Gemini LLM to translate complex data profiles into actionable, prioritized lifestyle interventions for users

Data Scientist (Summer Intern), Camping World – Good Sam, Chicago, IL, June 2023 – Aug 2023

- Built a K-modes clustering segmentation on 10k+ RV inventory records to drive targeted advertising and personalized outreach; clusters achieved ~85% internal similarity and were adopted for regional marketing campaigns
- Validated scalable ETL pipelines (SQL, Snowflake, SSMS), improving data quality and ensuring consistent features for downstream ML models
- Delivered a real-time Power BI dashboard with custom DAX metrics to track sales, inventory health and model KPIs for ~30 executives

Data Scientist, Unosis IT Solutions Private Limited, India, July 2019 – May 2022

- Developed and productionized a real-time CTR/CVR prediction pipeline (PySpark, feature store, LightGBM, served on SageMaker) using behavioral, session and product-features, improving offline AUC by ~0.08 and contributed to a measurable increase in ad clicks and revenue after rollout
- Developed end-to-end predictive models for churn, purchase intent and CLV on an e-commerce platform with 10M+ users; collaborated with product and ops to translate model outputs into targeted retention campaigns
- Applied BERT-based NLP to search and session logs to improve search relevance and query understanding recommending ranking changes that improved result relevance for long-tail queries
- Implemented ARIMA and Prophet forecasting pipelines for demand planning and dynamic pricing inputs, integrated forecasts into the enterprise DW for downstream analytics
- Led experiment design and causal evaluation for new recommendation features (A/B tests, power analysis, significance testing, guardrails)
- Built and optimized complex SQL and ETL processes, designed dimensional models (star schemas) to support analytics and ML feature extraction at scale
- Implemented MLOps pipelines using Kubernetes, Jenkins and AWS Lambda for model deployment and CI/CD, tracked experiments and model versions in MLFlow
- Developed Power BI dashboards and monitoring tools to translate complex technical and performance metrics into clear insights for engineering, product, and business stakeholders
- Collaborated cross-functionally with engineering teams and leadership to translate business requirements into actionable recommendations for improved user experience and business outcomes

Software Engineering Intern, Cojag Smart Technology, India, June 2018 – Feb 2019

- Contributed to full-stack web features and responsive UI design, integrating front-end components with backend services and MySQL.
- Implemented an IoT-enabled automated food ordering solution (Python + MySQL) to streamline order flow and delivery tracking.

EDUCATION

- University of Massachusetts Dartmouth, MA, USA – Master of Data Science, Sep 2022 – May 2025
- G.H. Raisoni College of Engineering, Nagpur, India – Bachelor of Engineering - Computer Science, Aug 2015 – May 2019

FEATURED PROJECTS

RAG System - Document Summarization and Self-Corrected Question Answering - GitHub

- Document Summarization & Self-Correcting Q&A (Gradio UI): production-grade RAG pipeline with response scoring and guardrail agent for quality control

Multimodal Medical Vision Question Answer with LLaVA Fine-Tuning - notebook

- Fine-tuned 7B vision and language model (LLaVA-1.5-7B) using Low-Rank Adaptation (LoRA) for radiology question answering, custom token-level F1 evaluation, and Hugging Face Hub deployment

American Express Kaggle Dataset: Insights into Fraudulent Transaction Detection - GitHub

- American Express Kaggle (5.5M rows): compared XGBoost, CatBoost, LSTM, and logistic baselines, CatBoost performed best in experiments

CIFAR100: Comparative Analysis of Deep Learning Architectures - GitHub

- Comparative analysis across 14 architectures (CNNs, EfficientNets, Vision Transformers, and Mobile Networks) for efficiency vs accuracy trade-offs, profiling training throughput, and inference latency

TEACHING & LEADERSHIP

Teaching Assistant – DSC 520, University of Massachusetts Dartmouth, MA, Sep 2024 – May 2025

- **High Performance Scientific Computing:** Mentored 75+ grads with implementing projects in **PyTorch**, Python optimization, and High-Performance Computing topics like **parallel processing**, **memory hierarchies**, **Linux command line**, and multi-core systems

Data Instructor, Digital Scholarship Hub, UMass Dartmouth, Dec 2023 – Aug 2024

- Led workshops on various tech topics like Data Visualization, ArcGIS, Machine Learning, SQL, and LangChain for faculty, students, and non-technical staff
- Developed Open Access guides for complex tools such as Git and data analysis, helping students and staff use them in their work
- Collaborated with the Scholarly Communication Librarian for research, reports, troubleshooting issues, maintaining digital reports, and auditing LibGuides

Teaching Assistant – CIS 190, University of Massachusetts Dartmouth, MA, Sep 2023 – Dec 2023

- **C Programming:** Led hands-on C programming labs, teaching key concepts like data types, variables, operators, and control structures to undergraduate students. Developed topic-specific labs, explained coding tasks, and graded assignments

CERTIFICATIONS & COMMUNITY

- AWS Certified Machine Learning Specialty – view
- Data Carpentries Certified Instructor (May 2025)
- Open-source maintainer: RIDE framework (docs & CLI), community-focused reproducible ML tooling

MASTERS THESIS, Jan 2023 – May 2025

Rapid Insight Data Engine: An open-source Python framework for analyzing Tabular Data - thesis

- **R.I.D.E.:** Built and maintain a no-code machine learning platform designed to help users implement a data analysis workflow to explore, clean, transform, and model data (CSV, Excel, and Parquet) backed by GPT-4 to analyze results and maintain memory for users to chat with data. Deployed using Docker and Kubernetes for scaling with users from my university - docs
- **RIDE-CLI:** Menu-driven CLI Python package that allows you to perform Feature Engineering on data (CSV, Excel, and Parquet), such as Data Quality checks, feature scaling and transformation, handle missing values, and perform regression and Classification through AutoML for streamlined data analysis workflows with 500+ downloads - docs